

GenAI-Powered Customer Support Quality Auditor

Project Statement:

This project aims to build a **GenAI-powered quality auditing platform** that reviews **customer support chats and calls**, assigns quality scores, detects compliance violations, and suggests improvements in real time. By combining **NLP, RAG pipelines, and speech-to-text transcription**, the platform leverages **LLMs** to evaluate tone, empathy, compliance with scripts, and resolution effectiveness. Designed for enterprises, BPOs, and SaaS companies, the solution enhances customer experience, ensures compliance, and reduces manual QA workload.

Outcomes:

- Automated analysis of chat and call transcripts.
- LLM-based scoring of quality metrics (empathy, resolution, compliance).
- Real-time detection of compliance breaches.
- RAG-powered contextual feedback and improvement suggestions.
- Dashboards for supervisors with agent-wise analytics.

Modules to be Implemented:

- **Data Ingestion & Transcription Layer**
Processes chat logs and transcribes calls using **Whisper/OpenAI STT**.
Normalizes transcripts into structured text.
- **Quality Scoring & Compliance Engine**
Uses **LLMs (GPT/Llama)** for scoring empathy, professionalism, compliance.
Flags violations and suggests improvements.
- **RAG Pipeline for Contextual Audits**
Integrates **LangChain + Pinecone/FAISS** for retrieval.
Provides context-aware auditing (e.g., checking against policy documents).
- **Dashboard & Visualization Hub**
Built with **Streamlit/React** for real-time analytics.
Displays agent-wise performance, compliance trends, and quality scores.

- **Alerting & Reporting Module**
Sends alerts for critical compliance issues.
Exports reports in PDF/Excel for supervisors.

Milestones:

- **Milestone 1: Weeks 1–2 Setup & Transcription**
Objective: Process chats and convert audio calls to text.
Tasks: Integrate Whisper/OpenAI STT, clean sample transcripts, define schema.
- **Milestone 2: Weeks 3–4 LLM Scoring Engine**
Objective: Build scoring pipeline for support quality.
Tasks: Apply GPT/Llama for scoring empathy, compliance, and issue resolution.
- **Milestone 3: Weeks 5–6 RAG & Dashboards**
Objective: Add contextual retrieval and supervisor dashboards.
Tasks: Integrate LangChain + Pinecone, build Streamlit/React dashboards.
- **Milestone 4: Weeks 7–8 Alerts & Deployment**
Objective: Enable alerts and deploy platform.
Tasks: Implement compliance alerts, enable PDF/Excel reports, deploy system.

Evaluation Criteria:

- **Milestone 1 Evaluation (Week 2):** Chats/calls ingested; transcription working.
- **Milestone 2 Evaluation (Week 4):** LLM scoring functional; quality scores generated.
- **Milestone 3 Evaluation (Week 6):** RAG-powered audits and dashboards operational.
- **Milestone 4 Evaluation (Week 8):** Alerts/reporting deployed; system live.